42. Second heart sound produced by closure of which valve? (ECE)

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Final CVs

1. **(Book Q.) A 3-year-old child is developing normally. Physical examination reveals a low-pitched cardiac murmur. An echocardiogram shows the presence of an ostium secundum, with a 1-cm defect. Which of the following abnormalities is most likely to be found in this child?**

A) Cyanosis at rest
B) Left-to-right shunt ✅
C) Mural thrombosis
D) Pericardial effusion

2. (Book Q.) Which of the following is the most common adverse effect of antihyperlipidemic drug therapy?

A) Elevated blood pressure
B) Gastrointestinal disturbance ✅
C) Neurologic problems
D) Heart palpitations

3. Which of the following contains sinusoidal capillaries?

A) Bone Marrow ✅
B) Skin

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C) Brain
D) Muscles

4. What is the normal coronary blood flow during rest?

A) 80 ml/min
B) 110 ml/min
C) 250 ml/min ✅
D) 50 ml/min

5. What is the mean arterial blood pressure (MAPB) if the systolic pressure was 130 mmHg and the diastolic pressure was 100 mmHg?

A) 110 mmHg ✅

6. Which of the following mediators causes vasodilation?

A. Nitric Oxide ✅

7. Which of the following changes the capillary filtering pressure?

A. Increased venous pressure ✅
B. Increase osmotic pressure
C. Decreased hydrostatic pressure

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8. Which of the following is considered to be the first step in the pathogenesis of Atherosclerosis?

A) Endothelial Dysfunction ✅

9. A 41-year old diabetic male has damaged vascular endothelium. Advanced glycation end products (AGEs) are present in this patient. Which of the following could result from such findings?

A) Blood coagulopathy ✅
B) Recurrent infections

10. In a normal ECG, which of the following coincides with Absolute Refractory Period of a myocardial cell?

A) Start of Q to top of T wave ✅

11. A patient presents to the em n ergency room with severe chest pain. Blood pressure is 80/55 and heartbeats are 120 bpm. Elevated jugular venous pressure and final bilateral crackles is noted. ECG results show elevated ST

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segments from leads II, III, and avF. Which of the following is most likely seen in this patient?

A) Infarct in the inferior part of the heart with congestive heart failure ✅
B) Infarct in the right part of the heart with hypotension
C) Infarct in the left part of the heart with bradycardia
D) Infarct in the inferior part of the heart with normal ventricular function

12. (case) A patient with left ventricular hypertrophy and increased afterload. Which of the cardiac cycles is most likely affected by this pathological finding?

A) Ejection phase
B) Isometric Contraction phase
C) Isometric Relaxation phase ✅

13. Which of the following heart sounds can be heard as a result of rapid blood filling from the atria into the ventricles (gushing)?

A) First heart sound

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B) Second heart sound
C) Third heart sound ✅
D) Fourth heart sound

14. The coronary sinus is derived from which of the following?

A) Sinus Venosus ✅

15. The imaging technique used to follow up a patient with aortic valve stenosis is:

A) MRI
B) Angiography
C) Echocardiography ✅
D) X-ray of skull

16. Which of the following should be monitored carefully when administering IV antiplatelet therapy?

A) BUN and Creatine
B) Liver Function Test

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C) Intracranial Pressure
D) Hemoglobin ✅

17. Which of the following primitive structures develops into the left auricle?

A) bulbus cordis
B) primitive atrium ✅
C) pulmonary veins
D) sinus venosus

18. Which structure can cause an Atrial Septal Defect (ASD) through an opening?

A) Foramen ovale ✅
B) Patent ductus venosus
C) Patent ductus arteriosus

19. A 29-year old male was involved in a car accident. He suffered from severe blood loss. He was tachycardic,

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hypotensive, with clammy and sweaty hands. Which of the following is most likely the type of shock?

A) Cardiogenic
B) Septic
C) Hypovolemic ✅
D) Distributive

20. Which of the following drugs can cause hemolytic anemia?

A) Methyldopa ✅
B) Clonidine

21. (case) What is the function of sterol regulatory element binding protein (SREBPs)?

A) Inhibit gene expression in cholesterol metabolism
B) activate gene expression in cholesterol and fatty acid metabolism ✅

22. Which of the following contraindicated to initiate beta blocker for congestive heart failure?

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A) Recent decompensated HF ✅
B) Ventricle ejection fraction < 40
C) Current digoxin use

23. A patient with MI showed a destruction of the cardiac muscle fibers. Which of the following cell junctions at intercalated disks and serve for keeping the integrity of the cardiac muscle during contraction are expected to be damaged in this condition?

A) Fascia adherens
B) Zonula occudens
C) Macula adherens ✅
D) Gap junctions

24. Which of the following intercalated discs play a role in spreading ionic current between myocytes?

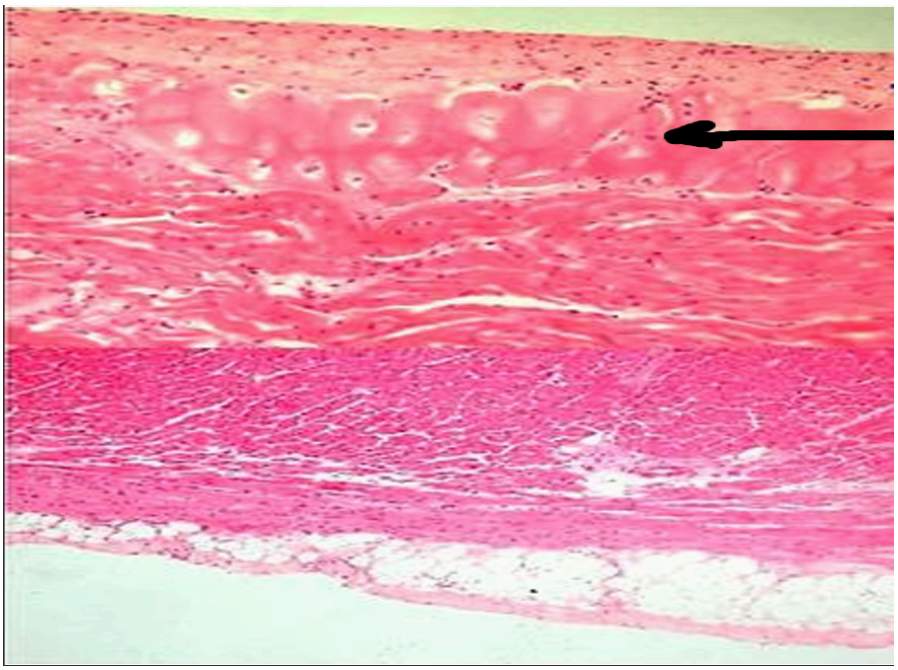
A) Gap junctions ✅
B) Fascia adherens
C) Zonula occudens
D) Macula adherens

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25. Purkinje fibers are present in which layer of the heart?

A) Subendocardial layer ان ✅



26. The indicated arrow shows which of the following? (should be endocardium but idk)

A) Myocardium
B) Epicardium
C) Fibrous cardiac skeleton

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D) Conductive cardiac system ✅

27. What are giant arteritis cells?

A) Necrotizing vasculitis
B) Lymphatic infiltration
C) Macrophages
D) Granulomatous inflammations ✅

28. A woman presents to the clinic with a painful mass under her fingernail. Biopsy of the mass shows modified smooth muscle cells with stroma. What's the diagnosis?

A) Glomus tumor ✅

29. What is a secondary cause of Hyperlipoproteinemia?

A) malnutrition
B) Hyperthyroidism
C) Nephrotic syndrome ✅

- **30. What causes hemopericardium and cardiac tamponade?**

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A) papillary muscle
B) dysfunction.
C) Free ventricular wall rupture ✅
D) Interventricular septum rupture

31. What is found directly behind the base of the heart?

A) esophagus ✅
B) Trachea
C) Ascending aorta
D) Traverse sinus

32. What's anterior to the oblique sinus?

A) right atrium
B) Left atrium ✅

33. Which of the following is characteristic for Heart Failure? (ECE)

A) Unilateral coarse lung crackles
B) Elevated Jugular venous pressure ✅

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34. Which of the following is the most useful in differentiating dyspnea? (ECE)

A) Duration of Dyspnea
B) Frequency of Cough
C) Presence of Orthopnea ✅

35. Which of the following is the best definition for palpitation? (ECE)

A) Awareness of irregular heartbeat ✅

36. What is the most common heart condition that presents with an irregularly irregular rhythm? (ECE)

A) Atrial Flutter
B) Ventricular Tachycardia
C) Atrial Fibrillation ✅
D) Ventricular Fibrillation

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37. Which of the following is the most accurate description of Janeway lesions? (ECE)

A) Painful nodules in pulp of fingers
B) Not painful spots on palms and soles of feet ✅

38. 2)Smoking is an important element of social history. Which of the following is a consequence or complication of smoking? (ECE)

A) Obesity
B) Epilepsy
C) Premature coronary artery disease ✅

39.Which of the following will cause central cyanosis? (ECE)

A) Left ventricular heart failure
B) Arteriolar emboli
C) High Altitude ✅
D) Exposure to cold

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40. Someone came with dyspnea, a cough that produces frothy sputum with Wheezing. And Heard with stethoscope Crackles, what is the diagnosis? (ECE)

A) pulmonary edema ✅
B) asthma
C) pneumonia
D) congestive heart failure

41.The correct position in cardiovascular examination is?(ECE)

A) 45 degree ✅
B) 90 degree
C) standing position
D) supine

A)

43.A 25-year old male patient has a past history of rheumatic fever. His auscultatory findings of the heart showed a loud first heart sound and a mid-diastolic murmur at apex. Which of the following is the most likely diagnosis? (ECE)

A) Aortic regurgitation

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B) Aortic stenosis
C) mitral regurgitation
D) mitral stenosis ✅

44. Case with a patient that had splinter hemorrhage. Most likely diagnosis? (ECE)

A) Infective Endocarditis ✅
B) Heart failure
C) Renal failure

45. A 52-year old man develops sudden severe central chest pain after lifting heavy cases while moving to a new house. The pain radiates to the back and both shoulders but not to either arm. His BP is 170/110 mmHg, pulse rate is 92 beats per minute and the ECG is normal. He is distressed and sweaty, but not nauseated. What would you cinder the most likely diagnosis? (ECE)

A) Myocardial Infarction
B) Aortic Dissection ✅

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C) Pulmonary Embolism
D) Pneumothorax

46. How is the SA node rate affected by sympathetic stimulation?

A) increase force of ventricular contraction
B) prolonged AV nodal dely
C) aortic blood pressure
D) increase slope of prepotential SA node ✅

47. Which of the following is directly proportional to the rate of blood flow?

A) Vessel radius ✅
B) Vessel length
C) Vessel viscosity
D) Hematocrit value

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48. A decrease in the slope of the prepotential SA node can be caused by?

A) Increased vagal stimulation ✅

49. What is the function of Atrio-Ventricular Bundle (AVB)?

A) Conducts impulses from SA node to ventricles ✅
B) Conducts single from the apex to the rest of the ventricle
C) ?
D) ?

50. Blockade of which artery will cut the supply to the apex of the heart?

A) Left circumflex
B) Right marginal
C) Posterior interventricular
D) Anterior interventricular ✅

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51. What is the benefit of SGLT2 inhibitors in patients with chronic heart failure?

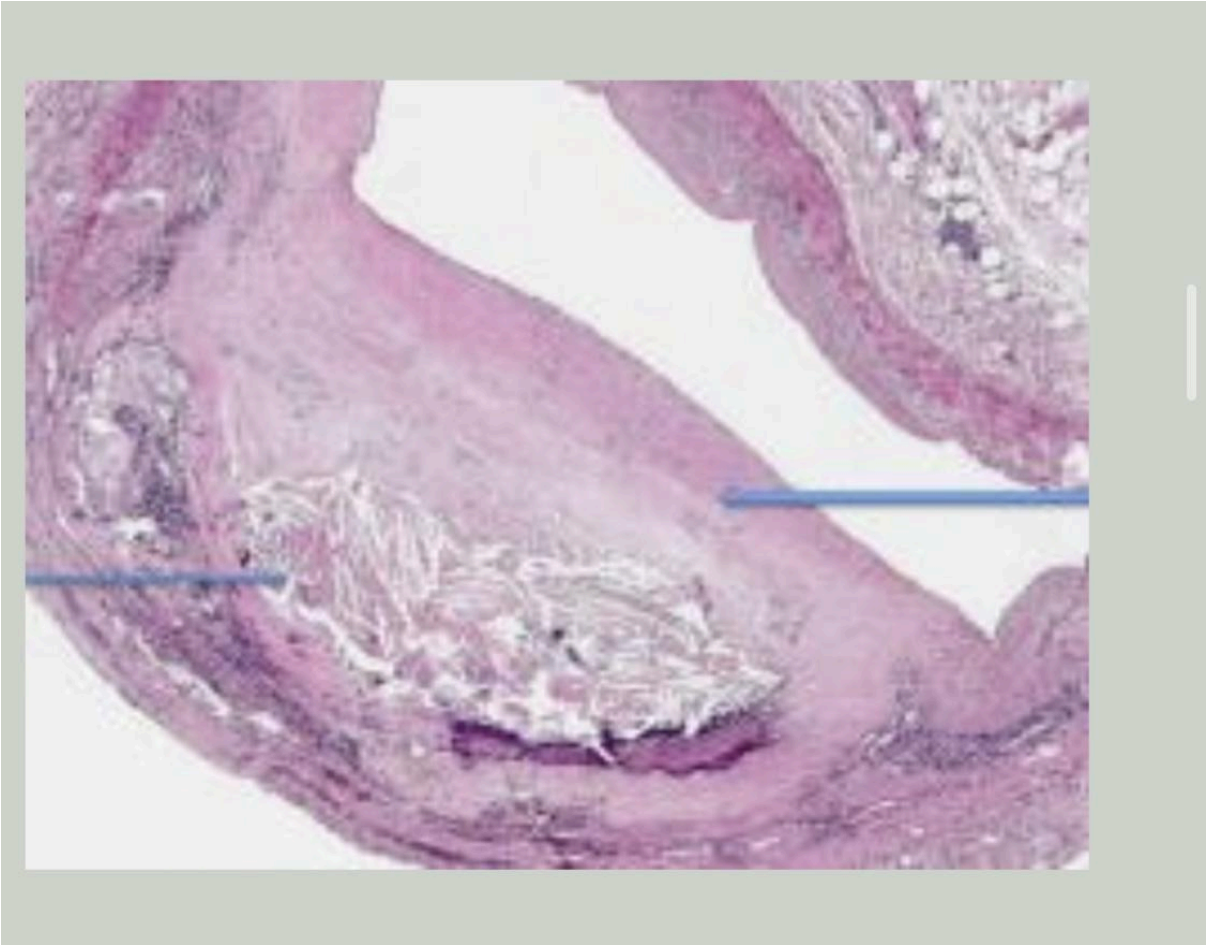
A) Reduced hospitalization for heart failure ✅

52. What primarily determines heart contractility?

A) HR
B) Preload (venous return) ✅

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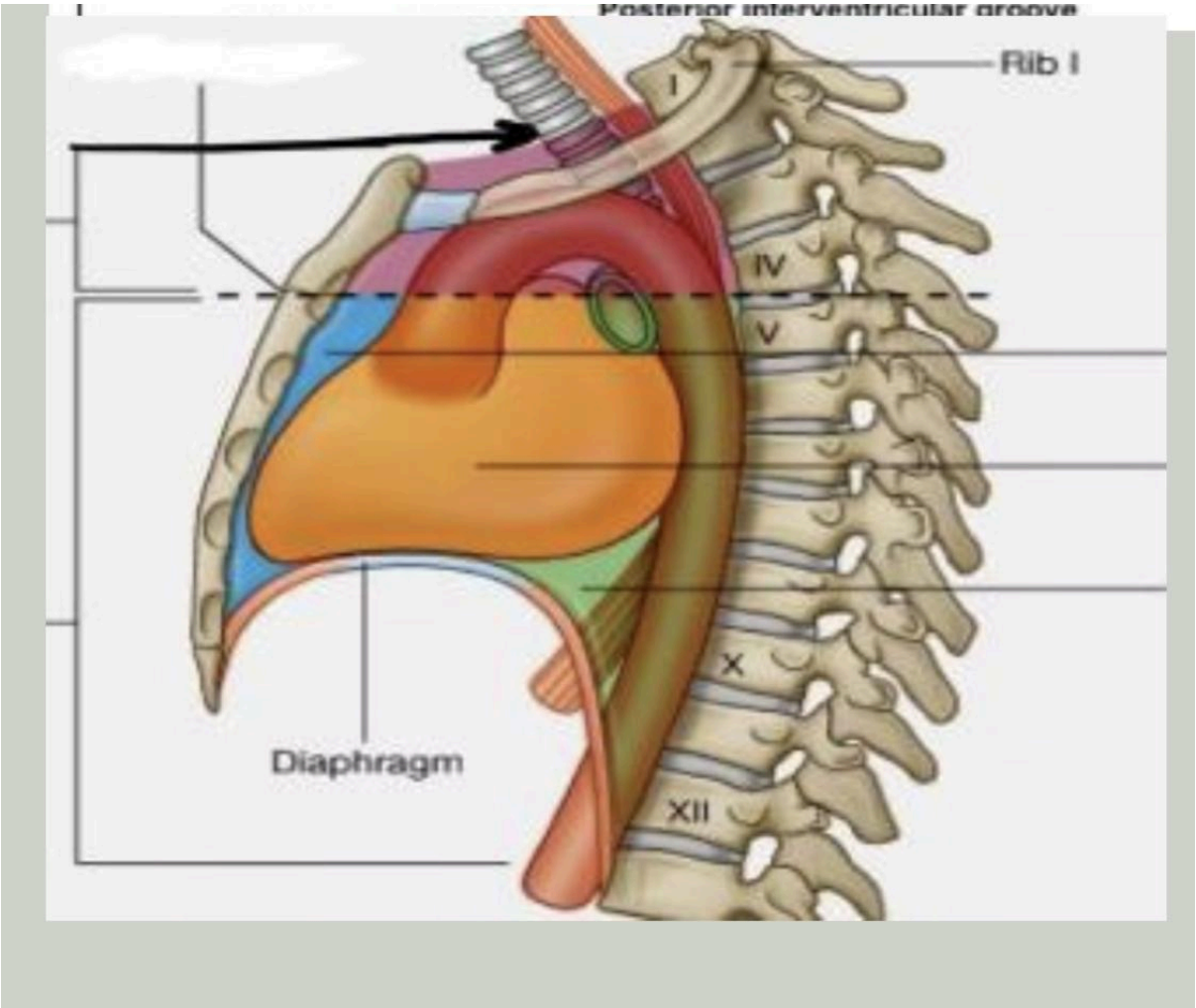
what is the type of fibrous cap in the following picture?

a- fibrous cap intact with central lipid core containing cholesterol ✅

b- fibrous cap intact with central necrosis granulmatous c- ulceration of its fibrous cap with hemorrhage

42. Second heart sound produced by closure of which valve? (ECE)

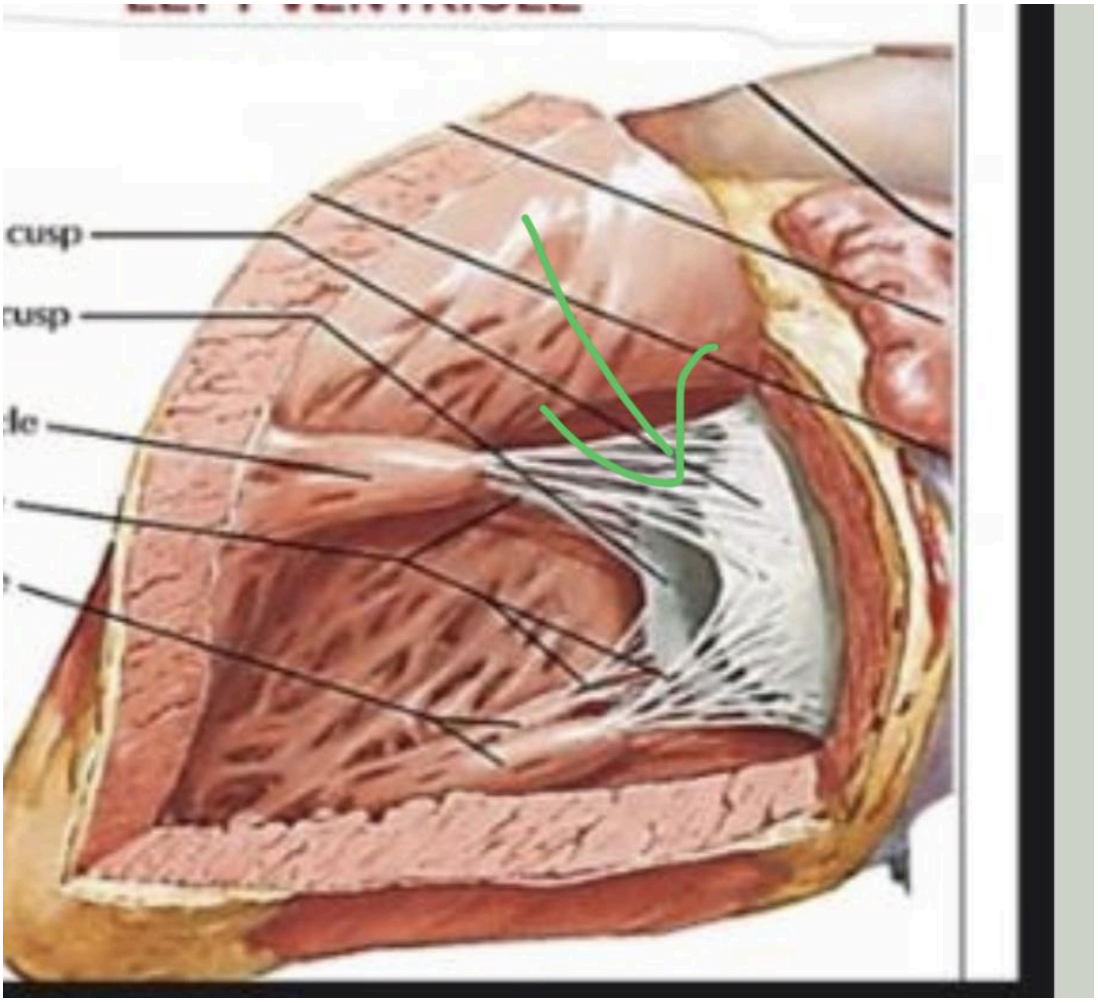
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A) t4-5

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(I think they asked about papillary muscle not chordea tendiona, yeah your right the arrow pointed at THE papillary muscle)

A) Chordea tendiona

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A case the ldl levels were normal but triglycerides were elevated.

A) mutation in ldl receptor
B) Lipoprotein lipase deficiency
C) Mutationin

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